# Daniel Hardesty Lewis

## Experience

### Professional

Apr. 2025 – Present **Co-founder**, *Homecastr*, New York, NY

Building a consumer real-estate pricing product focused on individual home price forecasting.

- Deployed a current-price prediction model for Manhattan condos and co-ops; achieved 12% MAPE on a holdout set and matched Zillow at 8.4% MAPE under an internal apples-to-apples evaluation protocol.
- Built a VAE-based latent representation layer over property, transaction, and neighborhood signals to stabilize learning and improve generalization under sparse and noisy features.
- Prototyped a diffusion-based forecasting backbone in Houston; measured 8% annualized compounding error on backtests spanning the 2020–2021 COVID period.
- Migrating market-specific models onto the diffusion backbone to standardize training, calibration, and evaluation across markets.
- Conducted investor discussions with a16z, MetaProp, and other firms; incorporated feedback into product positioning and milestones.
- Pitch deck: link

Nov. 2024 – Present **Co-founder**, *PoliBOM*, Seattle, WA

AI tariff mitigation and trade compliance product for manufacturers, centered on BOM analysis, retrieval, and policy scenario simulation

- Specified an agentic workflow and data contracts for BOM parsing, retrieval, and tariff simulation using Airflow, Elasticsearch and embeddings, Pydantic-validated JSON interfaces, and FastAPI serving backed by Supabase and Redis
- Built the tariff simulation and what-if framework for policy scenarios affecting BOM cost, lead times, and compliance risk, including China decoupling and Section 301 exposure analysis
- Presented the prototype to enterprise stakeholders including KPMG; market validation: KPMG subsequently launched a tariff-modeling product (link)
- Raising seed; running customer discovery with importers and exporters and iterating on distribution, pricing, and compliance workflows

Nov. 2023 – Present **Founder**, *Summit Geospatial*, New York, NY

High-resolution elevation data and web delivery for engineering and hazard workflows (summitgeospatial.com).

- Built a 0.5 m statewide seamless Texas DEM mosaic using nearest-neighbor stitching across 70+ state, federal, and local elevation sources.
- Built the web distribution platform end-to-end (data pipeline, tiling, hosting, delivery UX) to support engineering and planning use cases.
- Commercial discussions to license elevation datasets to research and AI organizations.

Aug. 2021 – Aug. 2023 **Senior Data Scientist & Technical Lead**, *Texas Advanced Computing Center (TACC)*, Austin, TX

- Spun out Summit Geospatial alongside UT's VPs of IP
- Led source-to-frontend data pipeline for $40M Texas Disaster Information System (TDIS) disaster resiliency initiative
- Delivered summitgeospatial.com, a precision decision-support platform used by civil engineers to quantify risk and allocate resources for disaster planning
- Contributed research towards Paola Passalacqua's 2022 EGU Bagnold Medal, one of geomorphology's highest honors
- Scaled climate and flood models on world-leading supercomputers, utilizing millions of dollars in computational resources and million-node jobs
- Evaluated and accepted tens of millions of dollars in compute requests
- Managed relationships and developed projects with federal, state agencies, and universities including NOAA, NSF, US Army Corps of Engineers, Texas General Land Office, USC, UT, A&M

Feb. 2018 – Jul. 2021 **Data Scientist & Research Engineer**, *Texas Advanced Computing Center (TACC)*, Austin, TX

- Competed in DARPA World Modelers program, focusing on disaster resiliency in East Africa
- Research received a mention from President of AAAI Yolanda Gil during her farewell speech
- Helped teach Petrobras deep learning on HPC

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## Research

May 2025 – Present **Research assistant to Dir. Electrical Engineering Zoran Kostic**, *Columbia University*, New York, NY
Targetting theory publications to DeepSeek's GRPO algorithm

Jan. 2025 – Present **Research assistant to Dir. Financial Engineering Ali Hirsa**, *Columbia University*, New York, NY
Achieved better than commercial $R^2$ on backtested holdout prediction of daily mutual fund returns on a variety of asset classes

- Using a DL latent factor model to produce better factors than our benchmark: Fama-French's 5-factor model
- Now targetting SHAP explainability publication

July 2019 – Aug. 2019 **Summer research intern to Pres. AAAI Yolanda Gil**, *University of Southern California*, Los Angeles, CA
Integrated hydrological models into team's DARPA-supported scientific model orchestration portal

## Teaching

Jan. 2018 – Dec. 2020 **Co-instructor**, *The University of Texas at Austin*, Austin, TX
Helped teach graduate-level courses: *Machine Learning for the Geosciences*, *Intelligent Systems for the Geosciences*, and *Scientific Computation*, including C++, CUDA, HPC, & optimization

# Education

expected 2026 **M.S., Urban Planning**, *Columbia University*, New York City, NY

2017 **B.S., Pure Mathematics**, *University of Texas*, Austin, TX

2017 **Certificate, Scientific Computation**, *University of Texas*, Austin, TX

# Key Projects

oppsAlert — Developed the first property- and owner-level ML model of NIMBYism, using 15+ years of 15000+ points of individual homeowner protest data provided by the City of Austin

Stealth, RegTech — *Seeking PE investment:* Teaming up with a top-proptech operator for a very large opportunity to aid cities achieve permitting compliance via simple pipeline collating publicly available imagery against permitting data

Flood Hazard Assessment — Partnered with the US Army Corps of Engineers to statistically model compound flood hazards in coastal Texas, integrating high-resolution topographic data with hydrological models.

Texas Disaster Information System (TDIS) — Led development of the core elevation data layer for a web-based spatial data system that supports resilient decision-making at the state and local levels.

Real-Time Inundation Mapping — Developed computationally efficient methods to produce high-resolution (1m) flood inundation maps from National Water Model outputs for emergency response personnel.

# Technology Skills

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| Programming Languages | Python, Bash, PL/pgSQL, Fortran, Perl, C++ | Operating Systems | GNU/Linux, macOS, Windows |
| Libraries & Frameworks | GDAL, CUDA, GeoPandas, RasterIO, NumPy, Plotly, Dash | Database Systems | PostgreSQL, Redis, Supabase, PostGIS |
| Scientific Software | MODFLOW, HAND, HEC-RAS, MESA | Cloud & Virtualization | AWS, Azure, Docker, Singularity |
| AI tooling | ChatGPT, Claude, Gemini, VSCode, v0.dev | ML | Representation Learning, VAE, Tree-based models |
| Clustering techniques & metrics | DBSCAN, Hierarchical, Silhouette score, Calinski-Harabasz, Davies-Bouldin | Statistical & ML metrics | $R^2$, MAPE, RMSE, KL-div, Reconstruction loss |

# Languages

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|---|---|---|---|
| Spanish | Native / Bilingual Proficiency | French | Professional Working Proficiency |

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